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// CSU CS 545
// Final Project Proposal

Forced Alignment Using Connectionist Temporal Classification

1. Overview

I am proposing a final project in which I train a bidirectional LSTM to perform forced alignment over spoken-word audio files from the TIMIT dataset. While we have covered LSTM's in class, the novelty comes from using an interesting loss function called Connectionist Temporal Classification (or CTC). The motivation for this idea comes from an interest in doing a prosody analysis and classification of speech using traditional audio engineering and music theory tools, which I was inspired to use as a way to reframe prosody information as "music". In turn, I realized the tokenization itself by finding word boundaries in speech was an interesting problem which fit the class and the requirements for this project quite well. The final deliverable is a notebook that explains CTC, how it works, how it differs from other loss methods like cross entropy, and why it's beneficial in forced alignment.

In researching this loss function, I discovered the paper which forms its foundation by Graves et. al. written in 2006 (Connectionist Temporal Classification: Labelling Unsegmented Sequence Data with Recurrent Neural Networks), in which the authors use the same data set that I have proposed here. As such, I gratefully plan to follow in their footsteps and aim to reproduce their findings with my own work.

2. CTC

The loss function itself is a way of measuring the alignment of an utterance with its transcript. The network will generate a matrix of probabilities, where the columns represent frames of audio, and the rows are all the possible sounds that could be perceived, including '_' for no sound at all. CTC is a way to analyze the entire matrix and extract the probability that all the columns (which are frames of audio) will condense down to something valid that matches the transcript. Or more precisely, the CTC returns the negative log of the summed probability of every path across the frames that condenses to the transcript. For example "c, c, c, a, a, a, t, t, t," and "c, _, _, a, _, _, t, _, _," both condense down to "c,a,t", and both would receive a low error score. The result, is a model which has classified at some accuracy, every frame of audio against a phoneme; given the transcript, a Viterbi search over that matrix then gives the most likely timing, and so where each word starts and ends.

3. biLSTM vs Transformer

The architecture decision to use a bidirectional LSTM comes from two main factors. Firstly, and perhaps most importantly, the TIMIT data set is large, but not gigantic. It contains about 10 utterances/sentences from 630 speakers, resulting in 6,300 documents total, which reaches about 5.5 hours of audio data if one were to listen continuously. While this is enough to train the model, and a huge amount for some poor souls to have labeled word boundaries by hand, its not as large as some of the competing spoken word corpuses which contain upwards on hundreds of hours of audio (the most important distinction being that those corpuses don't have consistently labeled word boundaries against an empirical/supervised metric). The size of the data set then limits how much tuning and training I can do which recommends a more reliable model rather than the potential for a more accurate one. The second reason for choosing a bidirectional LSTM other than reliability, is reproducibility and comparison. By following in the footsteps of Graves et. al. I can compare my findings to theirs directly.

With that said, the question arises if there is a better, newer, or more accurate architecture that would work for the same task. Since 2006 transformers have made a large jump forward. I want to propose a test

of this idea against the bidirectional model to see if the results could be improved on with such a different architecture. Not only will this allow me to choose the best results against the corpus for use in my next project (the "musical" prosody classification), but it will give me another set of results to look at critically/scientifically. If, for example, there is insufficient data in the TIMIT set to tune the transformer, I will still have usable data, and perhaps will still have made some interesting observations about how transformers behave in general. The transformer is a stretch goal, and I will consider the project complete without it if I have achieved a working CTC loss function and a demonstration through a model that provides meaningful output.

4. Timeline

My plan is to start out by loading some toy data into a python script and building the CTC loss function directly and checking it against PyTorch's built-in CTC loss. Adjusting and testing it until I can obtain reliable measurements from the toy data. Then I will consider the TIMIT data set, load it, clean it and prune it based on concepts we have covered thoroughly in homework assignments (duplication, normalization, encoding choices). This is the initial legwork that I need to have finished by early November. With a working loss function and clean data, I will then instantiate the model architecture and build the training loop, allowing any and all hyperparameters to be passed as arguments. The fun part, is then a controlled experiment over the data while modulating the hyperparameters to find where the best results are. Importantly, this step will also include a control group using cross entropy loss on TIMIT's frame-level labels (which CTC does not need) so as to provide a comparison for CTC. Both will be scored on phone error rate and on alignment boundary error (in milliseconds) against TIMIT's gold timings. I expect to be finished with this by Thanksgiving (November 26). Once I have collected some data that resembles my goal (a phone error rate near the ~30% from Graves et al.), I can build the transformer and integrate CTC into a more modern training loop. It's possible that the transformer will need more examples to be able to outperform the LSTM, so a negative result there is still a finding. The final goal is twofold: a working forced alignment, and, time permitting, a simple comparison between a reproduction of the biLSTM and a transformer on the same data with the same loss function.

After building the LSTM (and the transformer, time permitting) and training them both on the same data set I should have enough information to report my findings and complete the write up. I don't plan on working with a team.

Milestones:

- November 8: loss verified on toy data and TIMIT pipeline ready.
- November 26: BiLSTM with CTC versus cross entropy, scored on both metrics.
- November 30: core notebook drafted.
- Early December: final notebook, with the transformer comparison included if time permits.

Citations:

- https://www.cs.toronto.edu/~graves/icml_2006.pdf
- https://arxiv.org/abs/2209.08000
- Class notebooks.